Reviewer 1

Comments

Reviewer 1 - Initial submission, 17/10/2026
The paper probably contain valuable information and insights, however in its current form, it lacks a logical follow up and appropriate explanations to make the paper useful to the readers.

The abstract should start from explaining the field, and the context before delving into the high-level explanation of the approach. For now, the abstract is too technical. It is full of technical terms without defining what is the context, what we try to achieve and why, and what is measured and how.

For now, the paper lacks the necessary background and support for someone not familiar with this work to understand the paper. It seems this paper is a follow up to another paper, however the authors shouldn't assume the reader have already read and understood the previous paper. Many often, the explanations are anchored into the components or approaches already used by the authors in another paper. Some improvements or efforts to improve the paper are not explained. It seems that the authors wrote the paper rather for themselves as a list of to do tasks and improvements rather for an audience not familiar with their implementation. Many of the choices for different algorithms and approaches are not justified in this paper.

The writing lacks the logical follow up which would make the text much easier to understand. For example, in the introduction, the sentence "Sophia instead obtains the CTI target from an executed query over the Ariadne knowledge graph" comes in the third paragraph, however there is no previous explanation of what are Sophia or Ariadne? The text therefore is very difficult to understand. The explanation of Sophia comes in the section 2.1, however there is no previous explanation before to make this sentence in the introduction comprehensible.

Later on in the introduction section, the authors write "Two of the authors (Worth and Ahmad) are co-authors of AthenaBench (Alam et al., 30 2025); different data generation pipelines therefore do not establish organizationally independent evaluation". The authors need to provide some logical steps between the authorship and the evaluations, or a basic explanation what they're after in this sentence.

The authors should come up with a set of baseline methods, preferably the best in the field, then compare their work to those baseline works from experimental point of view. The table 1 is useful but only provides a theoretical comparison of their approach to the existing works.

In the section 4.1, the authors mention that Ariadne integrates 12 public CTI sources. However, some of these cited sources are just definitions, e.g., Sigma. The authors should be precise, do they mean a data base containing different Sigma rules? the same should be done for other cited sources.

Later in the section 4.1, the authors mention "These are approximate snapshot sizes, not a statement that all public CTI is covered.". For a scientific paper, it would be good to report the exact size of the graphs.

In the section 7.2, the results are provided but they are not interpreted: there is no explanation or intuition why sometimes the results are not good. The results are presented as they are without any effort to explain them.

In the section 7.5, the authors simply provide the examplles, by precising "they are illustrations, not a representative error sample". The authors need to provide more efforts to interpret and explain the examples and not limit themselves to simply detail the examples. The examples need to be explained and interpreted.

I am not sure I understand the rational behind the section 8: the authors provide the costs of running the approach. This may be beneficial for a funding proposal but for a scientific paper, as the costs change over time, perhaps in several months this evaluation is useless.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 22/10/2026
The manuscript presentes Glaukopis, a system designed to enhance cyber threat intelligence assistants through fine-tuning instructions based on knowledge graphs. The architecture utilizes a database called Ariadne and a template engine called Sophia to ensure that training data has clear and traceable provenance. Through a four-stage supervised training process, the Qwen2.5-32B base model demonstrates significant gains in specific security tasks, such as attack technique extraction and mitigation scoring. However, detailed evaluation of intermediate checkpoints reveals critical challenges, including catastrophic interference, collapses in assignment subtasks, and drops in overall reasoning performance. Finally, the study analyzes the impact of training steps, data alignment, and the computational costs associated with the local execution and deployment of specialized models.
While the submission possesses notable strengths—particularly its transparency regarding data provenance, auditability, and the self-correction of statistical discrepancies from earlier drafts—it currently suffers from fundamental methodological omissions, severe task interference, contamination loopholes, and missing bibliographic records.
The main issues that must be addressed are:
1. The central empirical claim surrounds a 4-stage SFT schedule. However, no single-stage (monolithic) multi-task control model was trained on an equivalent total token budget or update count. As the authors acknowledge, the study demonstrates "observable transitions with item-level uncertainty, not superiority over matched monolithic instruction tuning". Without a matched single-stage control, it is impossible to verify whether the 4-stage schedule provides a genuine learning advantage or merely introduces unnecessary complexity.
2. Hyperparameters vary concurrently across stage transitions. Hence, since dataset composition, sequence length, sample packing, and learning rates change simultaneously, stage transitions cannot isolate causal effects.
3. During Stage 3 (SOC drills), canonical attribution performance drops precipitously from 36.8 to 3.4 (a 33.5-point drop). Qualitative error taxonomy demonstrates that 124 out of 133 responses output a valid ATT&CK group name that is factually incorrect, and the model completely ceases generating standard G-codes.
4. In Vulnerability Severity Prediction (VSP), exact CVSS vector matching drops from 25.6% at Stage 2 down to 13.6% at the final checkpoint. Mismatches on Privileges

Required (PR) more than double (357 to 791 items), while Attack Vector (AV) and Scope (S) quadruples. Distance-based scoring masks this critical vulnerability judgment failure.
5. General reasoning on MMLU-Pro drops by 11.6 points for the 32B model (69.0 to 57.4) and up to 24.6 points on the MoE port. Performance gains are strictly isolated to AthenaBench—the benchmark aligned directly with the training schemas. External benchmarks like CyberSOCEval show zero net gain (42.6 vs. base 42.6).
6. Co-authors Worth and Ahmad are also co-authors of AthenaBench, the primary benchmark suite where performance gains are reported. The study is funded directly by Athena Security Group, LLC. Evaluating synthetic datasets built to match self-authored benchmark schemas without entity-disjoint, out-of-domain, or blind incident testing creates a substantial evaluation bias.
7. Tables 12 and 13 evaluate commercial hosted baselines including GPT-5.5, GPT-5.2, DeepSeek-V4-Pro, DeepSeek-V3.2-Exp, Gemini-3-Flash-Preview, and Gemini-2.5-Flash. Zero formal bibliographic references or technical report citations exist in the bibliography for these models. Full citations must be provided.